NAME: _____ Geometry (FL B.E.S.T.)

7.1 Polygon Angle Sums

Every convex polygon's interior angles can be found by splitting the polygon into triangles from one vertex: an n-gon splits into $(n-2)$ triangles, and since each triangle contributes $180°$, the interior angle sum is $S = (n-2)\cdot 180°$. Exterior angles (one per vertex, formed by extending each side) always add to exactly $360°$ for *any* convex polygon, no matter how many sides it has — walking once around the perimeter turns you through one full rotation. For a **regular** polygon (congruent sides and angles), split these totals evenly: each interior angle is $\frac{(n-2)180°}{n}$, and each exterior angle is $\frac{360°}{n}$. At any single vertex, the interior and exterior angles are supplementary. This worksheet goes beyond plugging into a formula — expect triangulation reasoning, two-variable algebra, and word problems that combine both formulas.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Find the interior angle sum of a convex heptagon ($n = 7$).

2. Find each interior angle of a regular decagon.

3. The interior angle sum is $2340°$. Find n.

4. Find each exterior angle of a regular 20-gon.

5. A hexagon has five angles of $120°$ each; find the sixth.

6. Find the exterior angle sum of a convex 100-gon.

7. A regular polygon has each exterior angle $24°$. Find n.

8. Find each interior angle of a regular pentagon.

9. A quadrilateral has angles $2x$, $3x$, $4x$, x. Find the largest angle.

10. Find the interior angle sum of a convex dodecagon ($n = 12$).

11. One exterior angle is $30°$. Find the interior angle at that same vertex.

12. A regular polygon has each interior angle $150°$. Find n.

13. A pentagon has angles $90°$, $100°$, $110°$, $120°$, and x. Find x.

14. A regular polygon's interior angle is exactly 8 times its exterior angle. Find the number of sides.

15. A convex nonagon has interior angles that are nine consecutive integers (in degrees). Find the smallest and largest angles.

✓ Answer Key

1. $900°$
2. $144°$
3. $n = 15$
4. $18°$
5. $120°$
6. $360°$
7. $n = 15$
8. $108°$
9. $144°$ ($x = 36°$)
10. $1800°$
11. $150°$
12. $n = 12$
13. $x = 120°$
14. $n = 18$ (each exterior $= 20°$, each interior $= 160°$)
15. Smallest $136°$, largest $144°$ (sum $= 1260°$)